

SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_MN_Duluth.Intl.AP-Duluth.ANGB.727450_TMYx.epw
- Building Name: SmallOffice
- Building Location: Duluth, MN
- Total Floor Area: 511 m²

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	window	Window upgrade: • Caulking application: polyurethane
Scenario 2	window	Window upgrade: • Caulking application: acrylic

Scenario 3	window	Window upgrade: • Caulking application: polyurethane • Secondary glazing application (skipped: SimpleGlazing windows)
Scenario 4	window	Window upgrade: • Secondary glazing application (skipped: SimpleGlazing windows)
Scenario 5	window	Window upgrade: • 2-pane replacement (skipped: SimpleGlazing windows) • Weatherstrip application (skipped: no OperableWindow) • Glazing film application (skipped: SimpleGlazing windows) • Caulking application: polyurethane • Window frame replacement (skipped) • Secondary glazing application (skipped: SimpleGlazing windows)

Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	236.54	236.54	0.00	0.00%
Scenario 2	Total Site Energy (GJ)	236.54	236.54	0.00	0.00%
Scenario 3	Total Site Energy (GJ)	236.54	236.54	0.00	0.00%
Scenario 4	Total Site Energy (GJ)	236.54	236.54	0.00	0.00%
Scenario 5	Total Site Energy (GJ)	236.54	232.16	4.38	1.85%

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison

Baseline		\$9,841
Scenario 5		\$9,730

■ Baseline ■ Scenario 5

Max saving: **\$-111 (-1.13%)**

Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr) Comparison

Baseline		20,894
Scenario 5		20,614

■ Baseline ■ Scenario 5

Max saving: **-280 kg CO₂e (-1.34%)**

Min Retrofit Construction Cost

\$0.00

Scenario 4

Min Retrofit Embodied Carbon

0.00 kg CO₂e

Embodied carbon intensity (ECI): 0.00 kgCO₂e/m²

Scenario 2

Lowest Retrofit Cost Payback Period

45 years

Scenario 5

Lowest Retrofit Carbon Payback Period

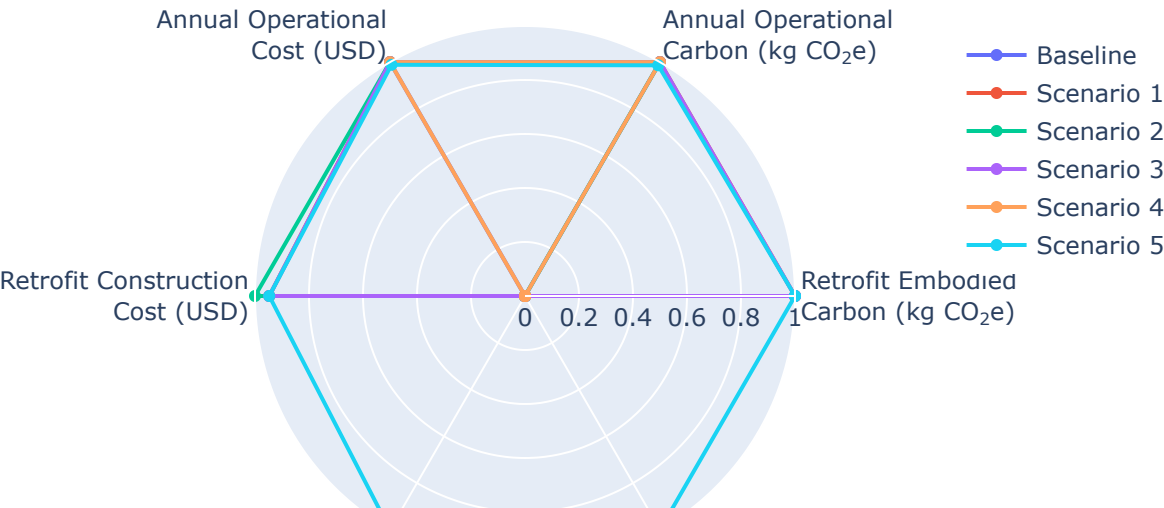
0.19 years

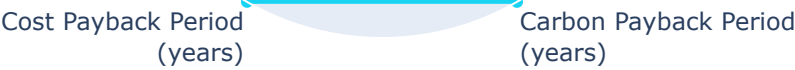
Scenario 5

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization

Parametric Scenario Comparison (Normalized Spider Chart)





Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Scenario 1		N/A
Scenario 2		N/A
Scenario 3		N/A
Scenario 4		N/A
Scenario 5		45 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr)).

Scenario 1		N/A
Scenario 2		N/A
Scenario 3		N/A
Scenario 4		N/A
Scenario 5		0.19 yrs

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

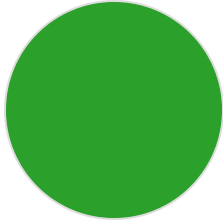
Scenario	Component	Material name	Volume (m ³)	Area (m ²)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m ³)	Product Lifetime (years)
Scenario 1	Window caulking	polyurethane	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 2	Window caulking	acrylic	0.0076	N/A	N/A	N/A	N/A	N/A	10
Scenario 3	Window caulking	polyurethane	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 4	Window caulking	N/A	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 5	Window caulking	polyurethane	0.0034	N/A	N/A	N/A	N/A	N/A	10

Result Summary

Per-scenario retrofit contribution breakdown. Each scenario includes two pie charts: cost and embodied carbon by retrofit measure.

Scenario 1

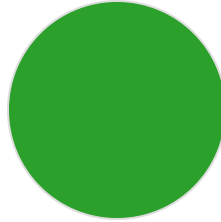
Cost breakdown by retrofit measure



■ Window: \$4,953 (100.0%)

Total: \$4,953

Carbon breakdown by retrofit measure

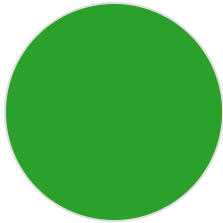


■ Window: 52 (100.0%)

Total: 52 kg CO₂e

Scenario 2

Cost breakdown by retrofit measure



■ Window: \$5,229 (100.0%)

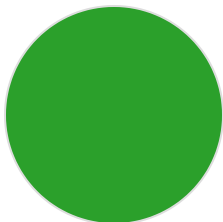
Total: \$5,229

Carbon breakdown by retrofit measure

No retrofit measure contribution data.

Scenario 3

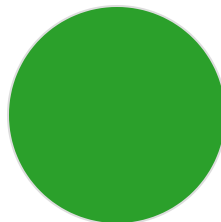
Cost breakdown by retrofit measure



■ Window: \$4,953 (100.0%)

Total: \$4,953

Carbon breakdown by retrofit measure



■ Window: 52 (100.0%)

Scenario 4

Cost breakdown by retrofit measure

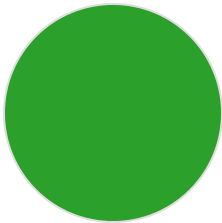
No retrofit measure contribution data.

Carbon breakdown by retrofit measure

No retrofit measure contribution data.

Scenario 5

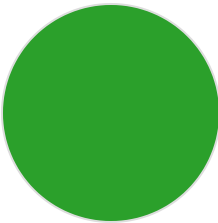
Cost breakdown by retrofit measure



■ Window: \$4,953 (100.0%)

Total: \$4,953

Carbon breakdown by retrofit measure



■ Window: 52 (100.0%)

Total: 52 kg CO₂e

Scenario	Retrofit Embodied Carbon (kg CO ₂ e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO ₂ e)	Annual Operational Cost (USD)
Baseline	0.00	\$0.00	20,894	\$9,841
Scenario 1	52	\$4,953	20,894	\$9,841
Scenario 2	0.00	\$5,229	20,894	\$9,841
Scenario 3	52	\$4,953	20,894	\$9,841
Scenario 4	0.00	\$0.00	20,894	\$9,841
Scenario 5	52	\$4,953	20,614	\$9,730

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